

International Economic Relations

ECON 308

Resources and Trade: The Heckscher-Ohlin Model
(Chapter 5)

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Introduction

- In addition to differences in labor productivity, trade occurs due to differences in resources across countries.
- The Heckscher-Ohlin theory argues that trade occurs due to differences in labor, labor skills, physical capital, capital, or other factors of production across countries.
 - Countries have different **relative abundance** of factors of production.
 - Production processes use factors of production with different **relative intensity**.

Two-Factor Heckscher-Ohlin Model

1. Two countries: home and foreign.
2. Two goods: cloth and food.
3. Two factors of production: labor and capital.
4. The mix of labor and capital used varies across goods.
5. The supply of labor and capital in each country is constant and varies across countries.
6. In the long run, both labor and capital can move across sectors, equalizing their returns (wage and rental rate) across sectors.

Production Possibilities (1 of 11)

- With more than one factor of production, the opportunity cost in production is no longer constant and the PPF is no longer a straight line. Why?
- Numerical example:

$K = 3000$, total amount of capital available for production

$L = 2000$, total amount of labor available for production

Production Possibilities (2 of 11)

- Suppose use a fixed mix of capital and labor in each sector.

$a_{KC} = 2$, capital used to produce one yard of cloth

$a_{LC} = 2$, labor used to produce one yard of cloth

$a_{KF} = 3$, capital used to produce one calorie of food

$a_{LF} = 1$, labor used to produce one calorie of food

Production Possibilities (3 of 11)

- Production possibilities describe different amounts of cloth and food that can be produced, given factor endowments and technology.
 - Q_C is total yards of cloth production
 - Q_F is total calories of food production

Production Possibilities (4 of 11)

- Production possibilities are influenced by **both** capital and labor:
 - Capital used to produce cloth and food cannot exceed the supply of capital available.

$$a_{KC}Q_C + a_{KF}Q_F \leq K$$

- Likewise, the labor used to produce cloth and food cannot exceed the supply of labor available.

$$a_{LC}Q_C + a_{LF}Q_F \leq L$$

Production Possibilities (5 of 11)

- Constraint on capital that capital used cannot exceed supply for the numerical example:

$$2Q_C + 3Q_F \leq 3000$$

- Constraint on labor that labor used cannot exceed labor supply:

$$2Q_C + Q_F \leq 2000$$

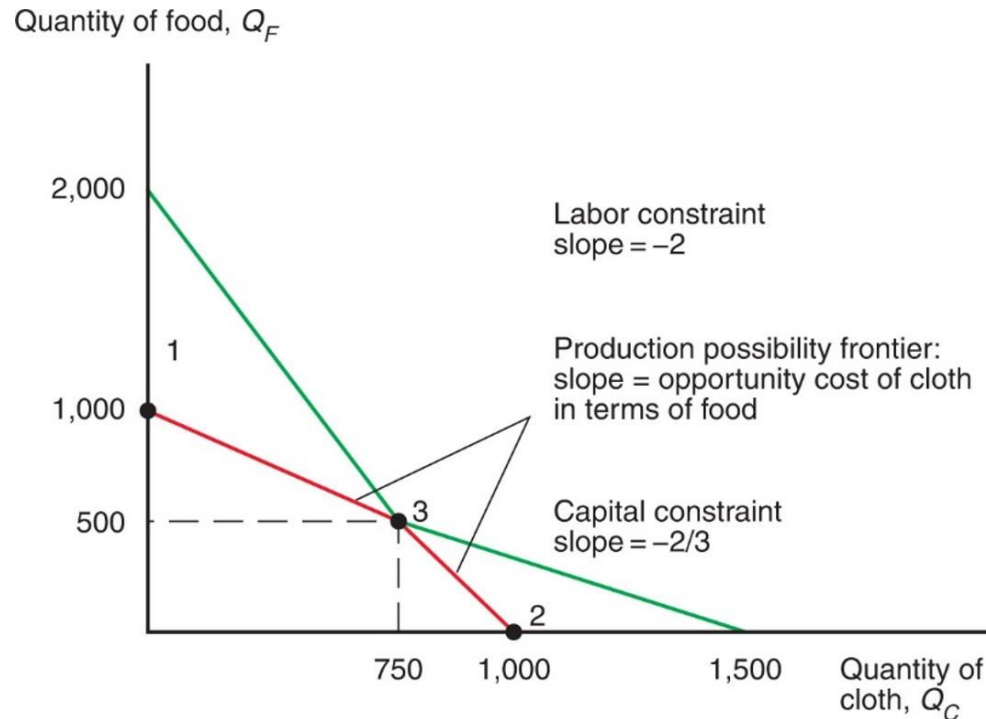
Production Possibilities (6 of 11)

- Economy must produce subject to both constraints—that is, it must have enough capital and labor.
- Without factor substitution, the production possibilities frontier is the interior of the two factor constraints.

Production Possibilities (7 of 11)

- Max food production 1000 (point 1) fully uses capital, with excess labor.
- Max cloth 1000 (point 2) fully uses labor, with excess capital.
- Intersection of labor and capital constraints occurs at 500 calories of food and 750 yards of cloth (point 3).

Figure 5.1 The Production Possibility Frontier without Factor Substitution



If capital cannot be substituted for labor or vice versa, the production possibility frontier in the factor-proportions model would be defined by two resource constraints: The economy can't use more than the available supply of labor (2,000 work-hours) or capital (3,000 machine-hours). So the production possibility frontier is defined by the red line in this figure.

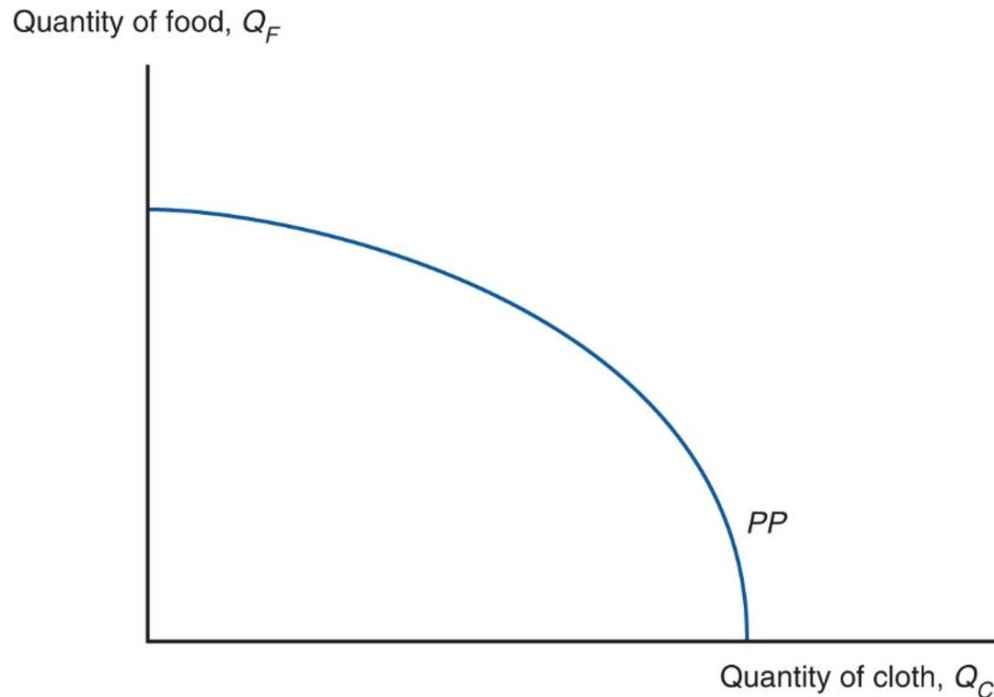
Production Possibilities (8 of 11)

- The opportunity cost of producing one more yard of cloth, in terms of food, is not constant:
 - **low** $\left(\frac{2}{3} \text{ in example}\right)$ when the economy produces a **low amount of cloth** and a high amount of food
 - **high** (2 in example) when the economy produces a **high amount of cloth** and a low amount of food
- Why? Because when the economy devotes more resources towards production of one good, the marginal productivity of those resources tends to be low so that the opportunity cost is high.

Production Possibilities (9 of 11)

- The above PPF equations do not allow substitution of capital for labor in production.
 - Unit factor requirements are constant along each line segment of the PPF.
- If producers can substitute one input for another in the production process, then the PPF is curved (bowed).
 - Opportunity cost of cloth increases as producers make more cloth.

Figure 5.2 The Production Possibility Frontier with Factor Substitution



If capital can be substituted for labor and vice versa, the production possibility frontier no longer has a kink. But it remains true that the opportunity cost of cloth in terms of food rises as the economy's production mix shifts toward cloth and away from food.

Production Possibilities (10 of 11)

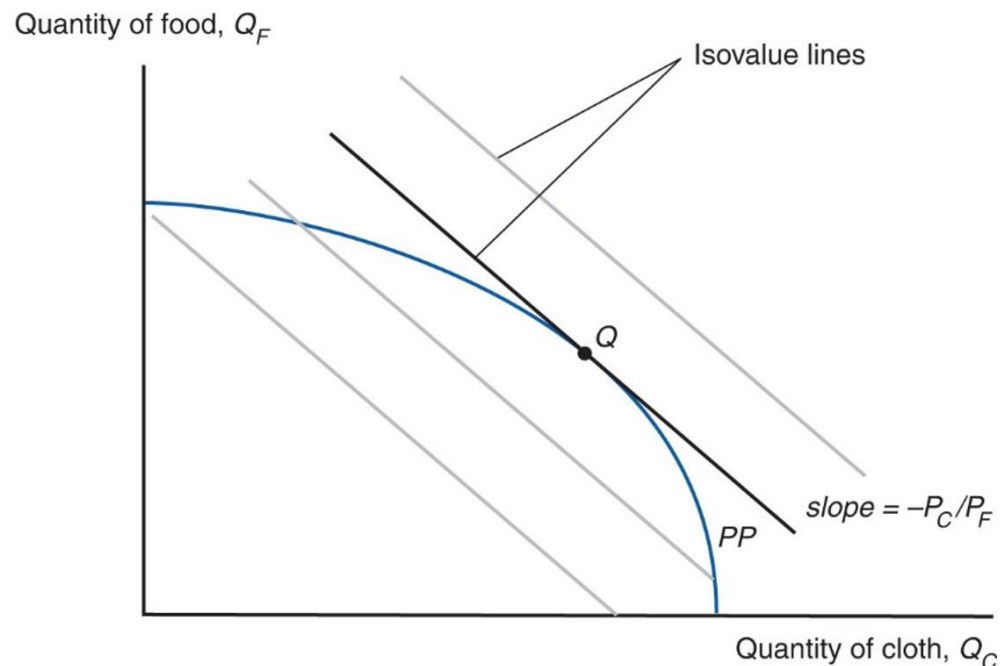
- What does the country produce?
- The economy produces at the point that maximizes the value of production, V .
- An **isovalue** line is a line representing a constant value of production, V :

$$V = P_C Q_C + P_F Q_F$$

— where P_C and P_F are the prices of cloth and food.

— slope of isovalue line is $-\frac{P_C}{P_F}$

Figure 5.3 Prices and Production



The economy produces at the point that maximizes the value of production given the prices it faces; this is the point on the highest possible isovalue line. At that point, the opportunity cost of cloth in

terms of food is equal to the relative price of cloth, $P_C > P_F$.

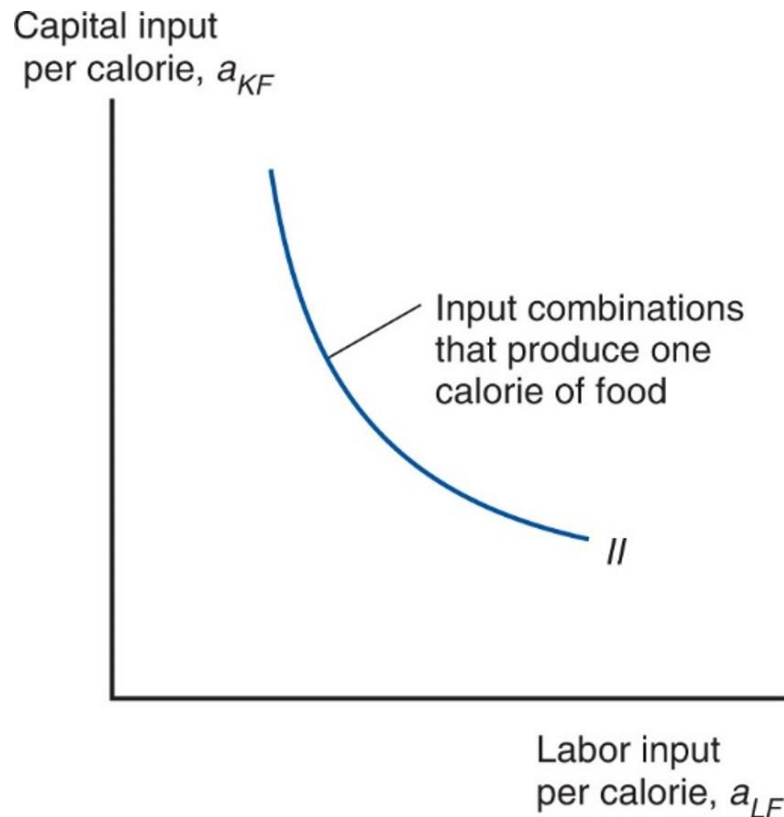
Production Possibilities (11 of 11)

- Given the relative price of cloth, the economy produces at the point Q that touches the highest possible isovalue line.
- At that point, the relative price of cloth equals the slope of the PPF, which equals **the opportunity cost of producing cloth**.
 - The trade-off in production equals the trade-off according to market prices.

Choosing the Mix of Inputs (1 of 2)

- Producers may choose different amounts of factors of production used to make cloth or food.
- Their choice depends on the wage, w , paid to labor and the rental rate, r , paid when renting capital.
- As the wage w increases relative to the rental rate r , producers use less labor and more capital in the production of both food and cloth.

Figure 5.4 Input Possibilities in Food Production



A farmer can produce a calorie of food with less capital if he or she uses more labor, and vice versa.

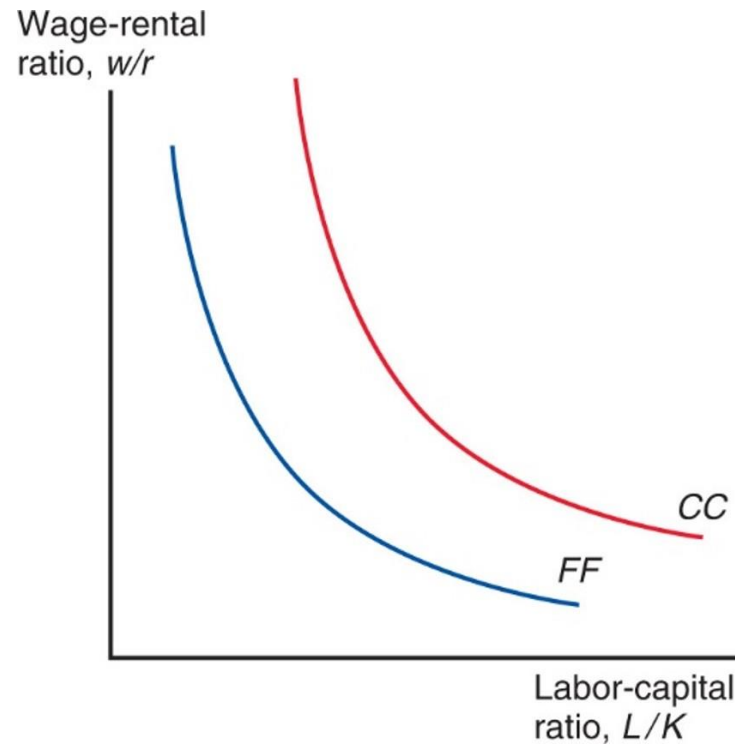
Choosing the Mix of Inputs (2 of 2)

- Assume that at any given factor prices, cloth production uses more labor relative to capital than food production uses:

$$\frac{a_{LC}}{a_{KC}} > \frac{a_{LF}}{a_{KF}} \quad \text{or} \quad \frac{L_C}{K_C} > \frac{L_F}{K_F}$$

- Production of cloth is relatively labor intensive, while production of food is relatively land intensive.
- Relative factor demand curve for cloth CC lies outside that for food FF .

Figure 5.5 Factor Prices and Input Choices



At any given wage-rental ratio, cloth production uses a higher labor-capital ratio; when this is the case, we say that cloth production is **labor-intensive** and that food production is **capital-intensive**.

Factor Prices and Goods Prices (1 of 3)

- In competitive markets, the price of a good should equal its cost of production, which depends on the factor prices.
- How changes in the wage and rent affect the cost of producing a good depends on the mix of factors used.
 - An increase in the rental rate of capital should affect the price of food more than the price of cloth since food is the capital intensive industry.
- Changes in $\frac{w}{r}$ are tied to changes in $\frac{P_C}{P_W}$.

Figure 5.6 Factor Prices and Goods Prices

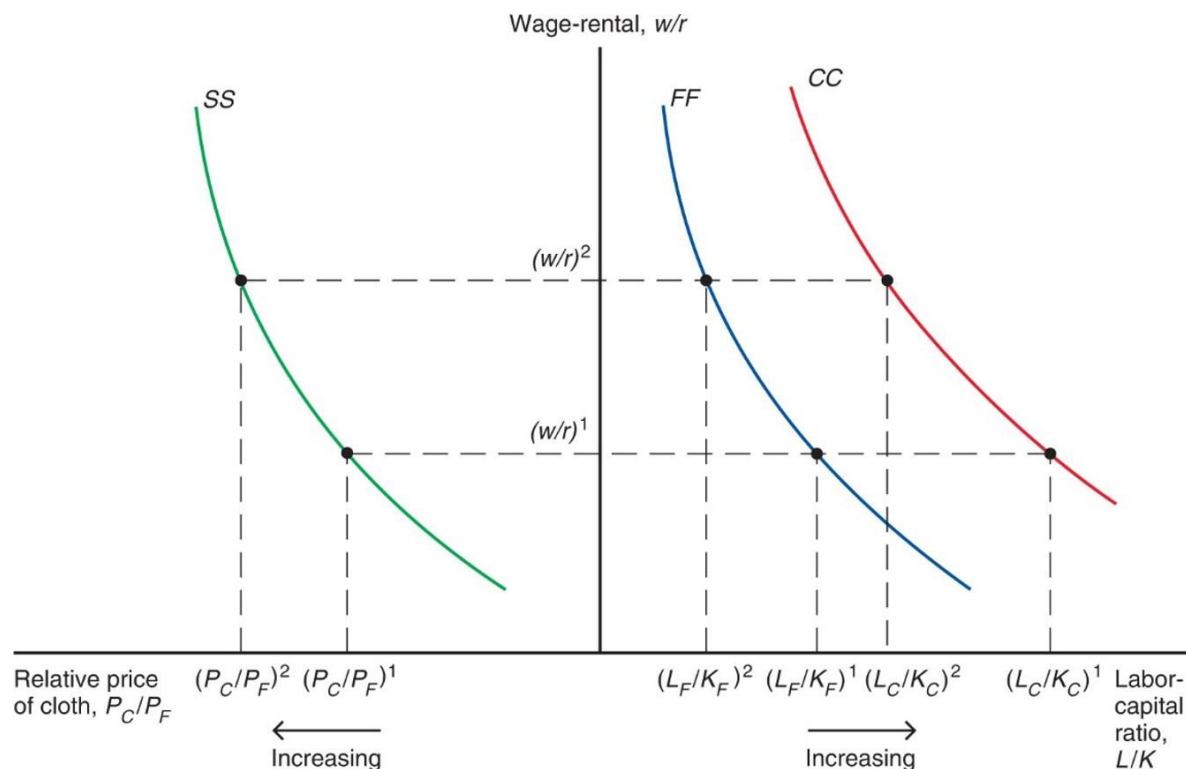


Because cloth production is labor-intensive while food production is capital-intensive, the higher the relative cost of labor, the higher must be the relative price of the labor-intensive good.

Factor Prices and Goods Prices (2 of 3)

- **Stolper-Samuelson theorem:** If the relative price of a good increases, then the real wage or rental rate of the factor used intensively in the production of that good increases, while the real wage or rental rate of the other factor decreases.
- Any change in the relative price of goods alters the distribution of income.

Figure 5.7 From Goods Prices to Input Choices



If the relative price of cloth rises, the wage-rental ratio must rise. This will cause the labor-capital ratio used in the production of both goods to drop.

Factor Prices and Goods Prices (3 of 3)

- An increase in the relative price of cloth, $\frac{P_C}{P_F}$, is predicted to
 - raise income of workers relative to that of capital owners, $\frac{w}{r}$.
 - raise the ratio of capital to labor services, $\frac{K}{L}$, used in both industries.
 - raise the real income (purchasing power) of workers and **lower the real income of capital owners**.

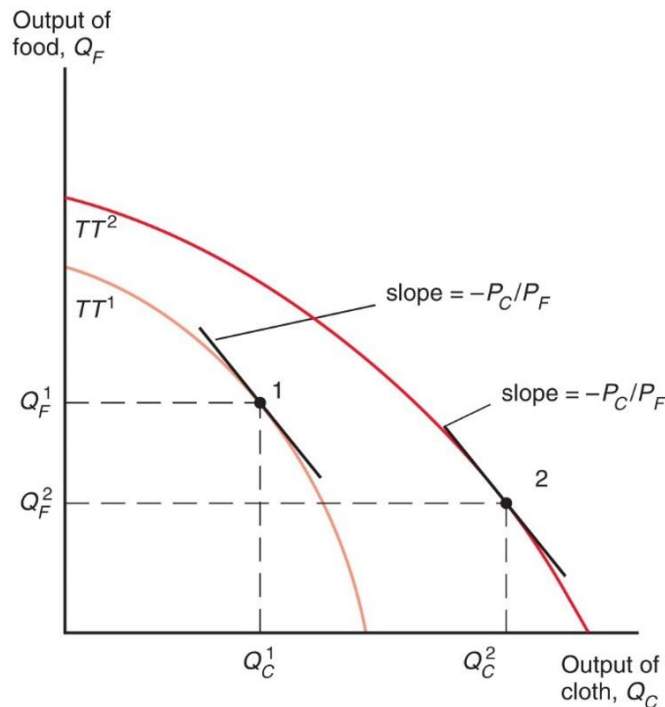
Resources and Output (1 of 3)

- How do levels of output change when the economy's resources change?
- **Rybczynski theorem:** If you hold output prices constant as the amount of a factor of production increases, then the supply of the good that uses this factor intensively increases and the supply of **the other good decreases**.

Resources and Output (2 of 3)

- Assume an economy's labor force grows, which implies that its ratio of labor to capital $\frac{L}{K}$ increases.
- Expansion of production possibilities is biased toward cloth.
- At a given relative price of cloth, the ratio of labor to capital used in both sectors remains constant.
- To employ the additional workers, the economy expands production of the relatively labor-intensive good cloth and contracts production of the relatively capital-intensive good food.

Figure 5.8 Resources and Production Possibilities



An increase in the supply of labor shifts the economy's production possibility frontier outward disproportionately in the direction of cloth production. At an unchanged relative price of cloth, food production declines.

Resources and Output (3 of 3)

- An economy with a **high ratio of labor to capital** produces a **high output of cloth relative to food**.
- Suppose that Home is **relatively abundant** in labor and Foreign in capital:

$$\frac{L}{K} > \frac{L^*}{K^*}$$

- Likewise, Home is **relatively scarce** in capital and Foreign in labor.
- Home will be relatively efficient at producing cloth because cloth is **relatively labor intensive**.

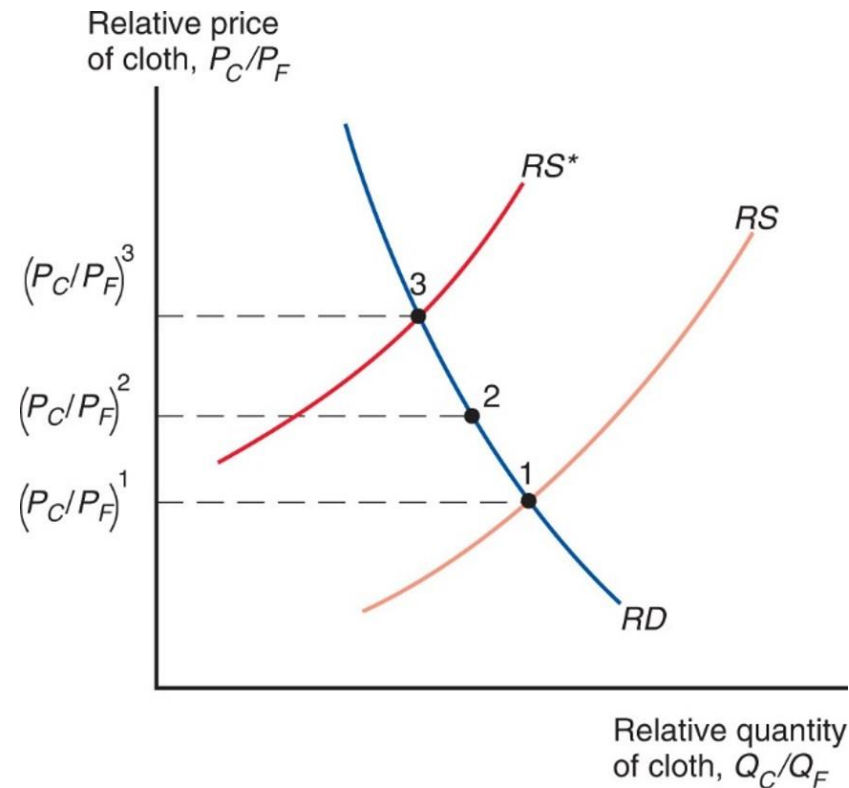
Trade in the Heckscher-Ohlin Model (1 of 5)

- The countries are assumed to have the same technology and the same tastes.
 - With the same technology, each economy has a comparative advantage in producing the good that relatively intensively uses the factors of production in which the country is relatively well endowed.
 - With the same tastes, the two countries will consume cloth to food in the same ratio when faced with the same relative price of cloth under free trade.

Trade in the Heckscher-Ohlin Model (2 of 5)

- Since cloth is relatively labor intensive, at each relative price of cloth to food, Home will produce a higher ratio of cloth to food than Foreign.
 - Home will have a larger relative supply of cloth to food than Foreign.
 - Home's relative supply curve lies to the right of Foreign's.

Figure 5.9 Trade Leads to a Convergence of Relative Prices



In the absence of trade, Home's equilibrium would be at point 1, where domestic relative supply RS intersects the relative demand curve RD . Similarly, Foreign's equilibrium would be at point 3. Trade leads to a world relative price that lies between the pretrade prices, such as at point 2.

Trade in the Heckscher-Ohlin Model (3 of 5)

- Like the Ricardian model, the Heckscher-Ohlin model predicts a convergence of relative prices with trade.
- With trade, the relative price of cloth rises in the relatively labor abundant (home) country and falls in the relatively labor scarce (foreign) country.

Trade in the Heckscher-Ohlin Model (4 of 5)

- Relative prices and the pattern of trade: In Home, the rise in the relative price of cloth leads to a rise in the relative production of cloth and a fall in relative consumption of cloth.
 - Home becomes an exporter of cloth and an importer of food.
- The decline in the relative price of cloth in Foreign leads it to become an importer of cloth and an exporter of food.

Trade in the Heckscher-Ohlin Model (5 of 5)

- **Heckscher-Ohlin theorem:** The country that is abundant in a factor exports the good whose production is intensive in that factor.
- This result generalizes to a correlation:
 - **Countries tend to export goods whose production is intensive in factors with which the countries are abundantly endowed.**

Trade and the Distribution of Income (1 of 5)

- Changes in relative prices can affect the earnings of labor and capital.
 - A rise in the price of cloth raises the purchasing power of labor in terms of both goods while lowering the purchasing power of capital in terms of both goods.
 - A rise in the price of food has the reverse effect.

Trade and the Distribution of Income (2 of 5)

- Thus, international trade can affect the distribution of income, even in the long run:
 - **Owners of a country's abundant factors gain from trade, but owners of a country's scarce factors lose.**
 - Factors of production that are used intensively by the import-competing industry are hurt by the opening of trade—regardless of the industry in which they are employed.

Trade and the Distribution of Income (3 of 5)

- Compared with the rest of the world, the United States is abundantly endowed with highly skilled labor while low-skilled labor is correspondingly scarce.
 - International trade has the potential to make low-skilled workers in the United States worse off—not just temporarily, but on a sustained basis.

Trade and the Distribution of Income (4 of 5)

- Changes in income distribution occur with every economic change, not only international trade.
 - Changes in technology, changes in consumer preferences, exhaustion of resources and discovery of new ones all affect income distribution.
 - Economists put most of the blame on technological change and the resulting premium paid on education as the major cause of increasing income inequality in the United States.
- It would be better to compensate the losers from trade (or any economic change) than prohibit trade.
 - The economy as a whole does benefit from trade.

Trade and the Distribution of Income (5 of 5)

- There is a political bias in trade politics: potential losers from trade are better politically organized than the winners from trade.
 - Losses are usually concentrated among a few, but gains are usually dispersed among many.
 - Each of you pays about \$8/year to restrict imports of sugar, and the total cost of this policy is about \$2 billion/year.
 - The benefits of this program total about \$1 billion, but this amount goes to relatively few sugar producers.

North-South Trade and Income Inequality (1 of 4)

- Over the last 40 years, countries such as South Korea, Mexico, and China have exported to the United States goods intensive in unskilled labor (e.g., clothing, shoes, toys, assembled goods).
- At the same time, income inequality has increased in the United States , as wages of unskilled workers have grown slowly compared to those of skilled workers.
- Did the former trend cause the latter trend?

North-South Trade and Income Inequality (2 of 4)

- The Heckscher-Ohlin model predicts that owners of relatively abundant factors will gain from trade and owners of relatively scarce factors will lose from trade.
 - Little evidence supporting this prediction exists.
- 1. According to the model, a change in the distribution of income occurs through changes in output prices, but there is no evidence of a change in the prices of skill-intensive goods relative to prices of unskilled-intensive goods.

North-South Trade and Income Inequality (3 of 4)

2. According to the model, wages of unskilled workers should increase in unskilled labor abundant countries relative to wages of skilled labor, but in some cases the reverse has occurred:
 - Wages of skilled labor have increased more rapidly in Mexico than wages of unskilled labor.
 - But compared to the United States and Canada, Mexico is supposed to be abundant in unskilled workers.

North-South Trade and Income Inequality (4 of 4)

3. Even if the model were exactly correct, trade is a small fraction of the U.S. economy, so its effects on U.S. prices and wages prices should be small.
- The majority view of trade economists is that the villain is not trade but rather new production technologies that put a greater emphasis on worker skills (such as the widespread introduction of computers and other advanced technologies in the workplace).

Skill-Biased Technological Change and Income Inequality (1 of 2)

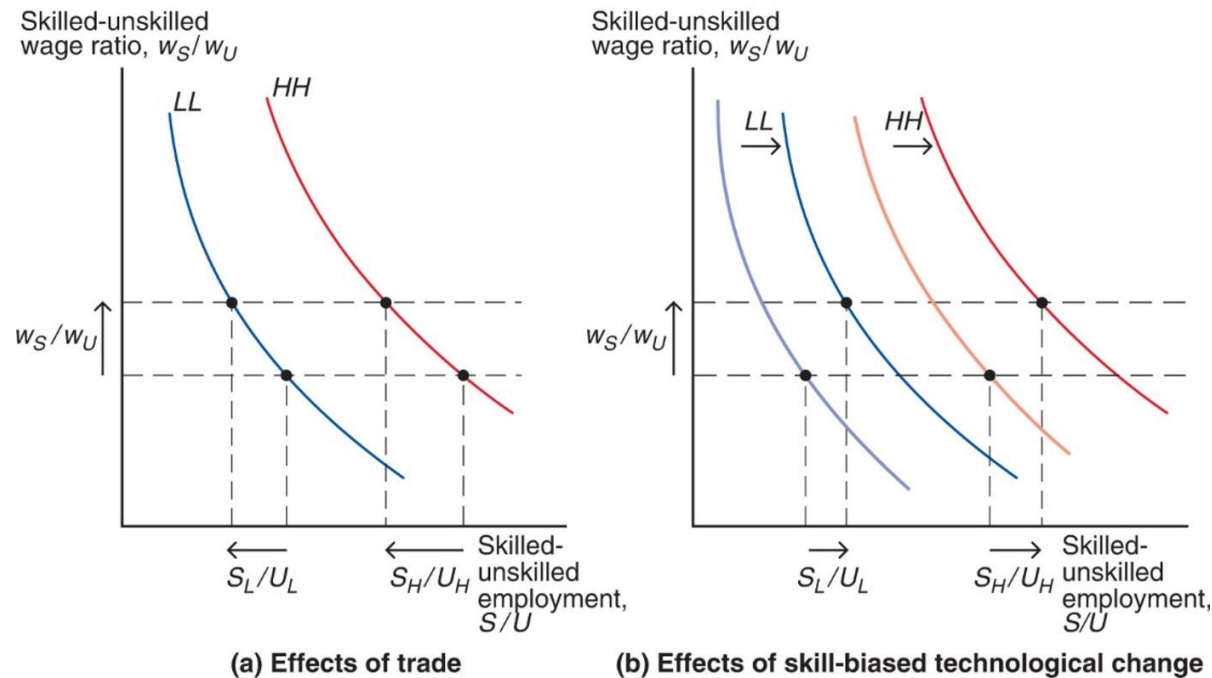
- Even though skilled labor becomes relatively more expensive, in panel (b) producers in both sectors respond to the skill-biased technological change by **increasing** their employment of skilled workers relative to unskilled workers.
 - The trade explanation in panel (a) predicts an opposite response for employment in both sectors.
- A widespread increase in the skilled labor ratios for most sectors in the U.S. economy points to the skill-biased technological explanation.

Skill-Biased Technological Change and Income Inequality

(2 of 2)

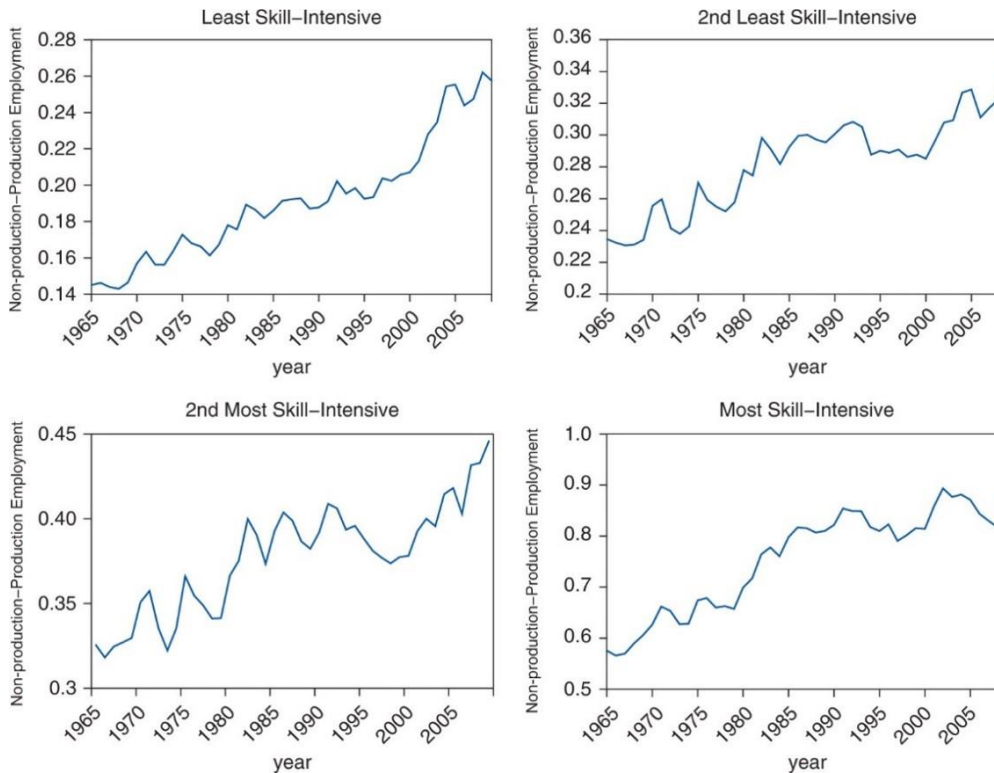
- Trade likely has been an indirect contributor to increases in wage inequality, by accelerating the process of technological change.
 - Firms that begin to export may upgrade to more skill-intensive production technologies.
 - Trade liberalization can then generate widespread technological change by inducing a large proportion of firms to make such technology-upgrade choices.
- Breaking up the production process across countries can increase the relative demand for skilled workers in developed countries similar to skill-biased technological change.

Figure 5.10 Increased Wage Inequality: Trade- or Skill-Biased Technological Change?



The LL and HH curves show the skilled-unskilled employment ratio as a function of the skilled-unskilled wage ratio in the low-tech and high-tech sectors. The high-tech sector is more skill-intensive than the low-tech sector, so the HH curve is shifted out relative to the LL curve.

Figure 5.11 Evolution of U.S. Non-Production–Production Employment Ratios in Four Groups of Sectors



Sectors are grouped based on their skill intensity. The non-production–production employment ratio has increased over time in all four sector groups.

Source: NBER-CES Manufacturing Productivity Database.

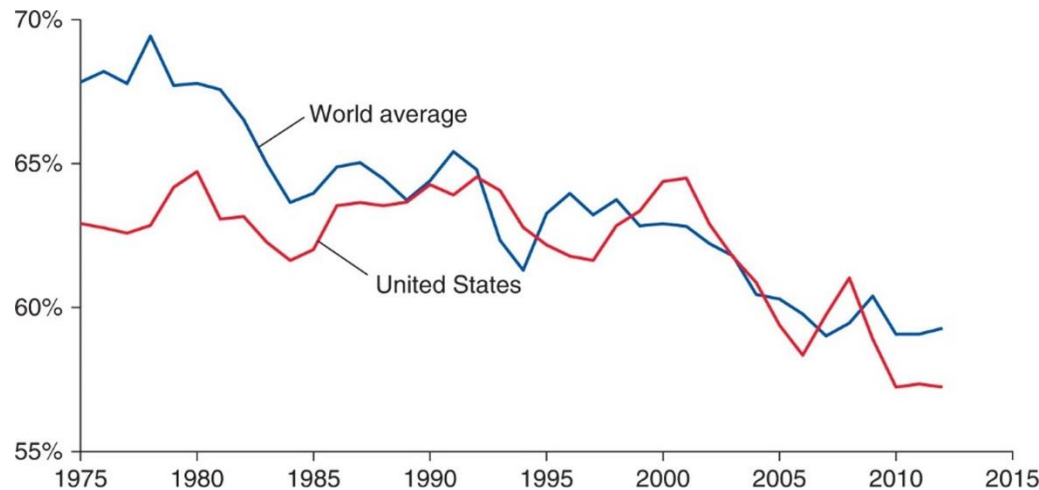
Capital-Skill Complementarity (1 of 2)

- Figure 5.12 shows that the trend toward lower labor income share (and higher capital shares) is a worldwide phenomenon.
 - Experienced in labor-abundant countries (including China, India, and Mexico) to the same extent as it has been for capital-abundant countries such as the United States.
- The evidence supports an explanation based on technological changes within sectors—not increased import competition.

Capital-Skill Complementarity (2 of 2)

- Technological change often takes the form of new and better machines (capital) that displace unskilled workers but still require skilled workers.
 - This generates higher returns for both capital and skilled workers while depressing the returns to unskilled workers.
- Increased automation can explain the observed worldwide increases in both wage inequality and the returns to capital, as well as the within-sector increases in the employment share of (relatively skilled) non-production workers.

Figure 5.12 U.S. and Average World Corporate Labor Share



Unweighted world average for all 59 countries with available data.

Source: Loukas Karabarbounis and Brent Neiman, “The Global Decline of the Labor Share,” **The Quarterly Journal of Economics** 129.1 (2014), pp. 61–103.

Factor Price Equalization (1 of 3)

- Unlike the Ricardian model, the Heckscher-Ohlin model predicts that factor prices will be equalized among countries that trade.
- Free trade equalizes relative output prices.
- Due to the connection between output prices and factor prices, factor prices are also equalized.
- Trade increases the demand of goods produced by relatively abundant factors, indirectly increasing the demand of these factors, raising the prices of the relatively abundant factors.

Factor Price Equalization (2 of 3)

- In the real world, factor prices are not equal across countries.
- The model assumes that trading countries produce the same goods, but countries may produce different goods if their factor ratios radically differ.
- The model also assumes that trading countries have the same technology, but different technologies could affect the productivities of factors and therefore the wages/rates paid to these factors.

Table 5.1 Comparative International Wage Rates

Country	Hourly Compensation of Manufacturing Workers 2016 (United States = 100)
Switzerland	154.64
Germany	110.63
United States	100.00
United Kingdom	72.79
Japan	67.80
South Korea	58.87
Argentina	42.97
Greece	40.22
Portugal	28.08
Czech Republic	27.43
Poland	21.83
Brazil	20.45
Turkey	15.61
Mexico	10.02
Philippines	5.27

Source: The Conference Board, International Labor Comparisons.

Factor Price Equalization (3 of 3)

- The model also ignores trade barriers and transportation costs, which may prevent output prices and thus factor prices from equalizing.
- The model predicts outcomes for the long run, but after an economy liberalizes trade, factors of production may not quickly move to the industries that intensively use abundant factors.
 - In the short run, the productivity of factors will be determined by their use in their current industry, so that their wage/rental rate may vary across countries.